亏格零曲线极小模型非唯一时的特殊纤维

Counting scope. The primary problem is Tag 0CDA, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题从亏格零曲线存在两个不同极小模型推出每个特殊纤维都是射影直线。证明比较两个模型的共同爆破，先收缩特殊纤维重数大于一的异常曲线降低爆破次数，再分析未收缩的异常曲线与原特殊纤维的同构。允许曲面双有理分解和异常曲线收缩后，重数分析、归纳及模型比较仍由本证明完成，判 H2。.

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1. Models

In this chapter R will be a discrete valuation ring and K will be its fraction field. If needed we will denote $\pi \in R$ a uniformizer and $k = R/(\pi)$ its residue field.

Let V be an algebraic K-scheme (Varieties, Definition 1.5). A *model* for V will mean a flat finite type[1] morphism $X \to \mathrm{Spec}(R)$ endowed with an isomorphism $V \to X_K = X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(K)$. We often will identify V and the generic fibre X_K of X and just write $V = X_K$. The special fibre is $X_k = X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$. A *morphism of models* $X \to X'$ *for* V is a morphism $X \to X'$ of schemes over R which induces the identity on V.

We will say X *is a proper model of* V if X is a model of V and the structure morphism $X \to \mathrm{Spec}(R)$ is proper. Similarly for separated models, smooth models, and add more here. We will say X *is a regular model of* V if X is a model of V and X is a regular scheme. Similarly for normal models, reduced models, and add more here.

Let $R \subset R'$ be an extension of discrete valuation rings (More on Algebra, Definition 1.6). This induces an extension K'/K of fraction fields. Given an algebraic scheme V over K, denote V' the base change $V \times_{\mathrm{Spec}(K)} \mathrm{Spec}(K')$. Then there is a functor

$$\text{models for } V \text{ over } R \longrightarrow \text{models for } V' \text{ over } R'$$

sending X to $X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(R')$.

Definition 1.1. Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$. A *minimal model* will be a regular, proper model X for C such that X does not contain an exceptional curve of the first kind (Resolution of Surfaces, Section section minus one (Tag 0C2I)).

Definition 1.2. In Situation 1.7 the *numerical type associated to* X is the numerical type described in Lemma lemma numerical type of model (Tag 0CA4).

Lemma 1.3. *In Situation 1.7. The following are equivalent*

[1] Occasionally it is useful to allow models to be locally of finite type over R, but we'll cross that bridge when we come to it.

(1) *X is a minimal model, and*
(2) *the numerical type associated to X is minimal.*

Lemma 1.4 (Genus zero)**.** *The only minimal numerical type of genus zero is $n = 1$, $m_1 = 1$, $a_{11} = 0$, $w_1 = 1$, $g_1 = 0$.*

证明. Follows from Lemmas lemma non irreducible minimal type genus at least one (Tag 0C7B) and lemma irreducible (Tag 0C73). □

Definition 1.5. Let k be a field. An *algebraic k-scheme* is a scheme X over k such that the structure morphism $X \to \mathrm{Spec}(k)$ is of finite type. A *locally algebraic k-scheme* is a scheme X over k such that the structure morphism $X \to \mathrm{Spec}(k)$ is locally of finite type.

Definition 1.6. We say that $A \to B$ or $A \subset B$ is an *extension of discrete valuation rings* if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or *residue degree* of the extension $A \subset B$.

Situation 1.7. Let R be a discrete valuation ring with fraction field K, residue field k, and uniformizer π. Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$. Let X be a regular proper model of C. Let $C_1, \ldots, C_n$ be the irreducible components of the special fibre X_k. Write $X_k = \sum m_i C_i$ as in Lemma lemma divisor special fiber (Tag 0C5Z).

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Lemma 1.8. *Let C be a smooth projective curve over K with $H^0(C, \mathcal{O}_C) = K$ and genus 0. If there is more than one minimal model for C, then the special fibre of every minimal model is isomorphic to $\mathbf{P}^1_k$.*

This lemma can be improved to say that the birational transformation between two nonisomorphic minimal models can be factored as a sequence of elementary transformations as in Example example nonunique in genus zero (Tag 0CA0). If we ever need this, we will precisely formulate and prove this here.

证明. Let X be some minimal model of C. The numerical type associated to X has genus 0 and is minimal (Definition 1.2 and Lemma 1.3). Hence by Lemma 1.4 we see that X_k is reduced, irreducible, has $H^0(X_k, \mathcal{O}_{X_k}) = k$, and has genus 0. Let Y be a second minimal model for C which is not isomorphic to X. By Resolution of Surfaces, Lemma lemma birational regular surfaces (Tag 0C5S) there exists a diagram of S-morphisms

$$X = X_0 \leftarrow X_1 \leftarrow \ldots \leftarrow X_n = Y_m \to \ldots \to Y_1 \to Y_0 = Y$$

where each morphism is a blowup in a closed point. We will prove the lemma by induction on m. The base case is $m = 0$; it is true in this case because we assumed that Y is minimal hence this would mean $n = 0$, but X is not isomorphic to Y, so this does not happen, i.e., there is nothing to check.

Before we continue, note that $n+1 = m+1$ is equal to the number of irreducible components of the special fibre of $X_n = Y_m$ because both X_k and Y_k are irreducible. Another observation we will use below is that if $X' \to X''$ is a morphism of regular proper models for C, then $X' \to X''$ is an isomorphism over an open set of X'' whose complement is a finite set of closed points of the special fibre of X'', see Varieties, Lemma lemma modification normal iso over codimension 1 (Tag 0BFP). In fact, any such $X' \to X''$ is a sequence of blowing ups in closed points (Resolution of Surfaces, Lemma lemma proper birational regular surfaces (Tag 0C5R)) and the number of blowups is the difference in the number of irreducible components of the special fibres of X' and X''.

Let $E_i \subset Y_i$, $m \geq i \geq 1$ be the curve which is contracted by the morphism $Y_i \to Y_{i-1}$. Let i be the biggest index such that E_i has multiplicity > 1 in the special fibre of Y_i. Then the further blowups $Y_m \to \ldots \to Y_{i+1} \to Y_i$ are isomorphisms over E_i since otherwise E_j for some $j > i$ would have multiplicity > 1. Let $E \subset Y_m$ be the inverse image of E_i. By what we just said $E \subset Y_m$ is an exceptional curve of the first kind. Let $Y_m \to Y'$ be the contraction of E (which exists by Resolution of Surfaces, Lemma lemma contract when quasi projective (Tag 0C2N)). The morphism $Y_m \to X$ has to contract E, because X_k is reduced. Hence there are morphisms $Y' \to Y$ and $Y' \to X$ (by Resolution of Surfaces, Lemma lemma factor through contraction (Tag 0C5J)) which are compositions of at most $n-1 = m-1$ contractions of exceptional curves (see discussion above). We win by induction on m. Upshot: we may assume that the special fibres of all of the curves X_i and Y_i are reduced.

Since the fibres of X_i and Y_i are reduced, it has to be the case that the blowups $X_i \to X_{i-1}$ and $Y_i \to Y_{i-1}$ happen in closed points which are regular points of the special fibres. Namely, if X'' is a regular model for C and if $x \in X''$ is a closed point of the special fibre, and $\pi \in \mathfrak{m}_x^2$, then the exceptional fibre E of the blowup $X' \to X''$ at x has multiplicity at least 2 in the special fibre of X' (local computation omitted). Hence $\mathcal{O}_{X''_k,x} = \mathcal{O}_{X'',x}/\pi$ is regular (Algebra, Lemma lemma regular ring CM (Tag 00NQ)) as claimed. In particular x is a Cartier divisor on the unique irreducible component Z' of X''_k it lies on (Varieties, Lemma lemma regular point on curve (Tag 0B8Y)). It follows that the strict transform $Z \subset X'$ of Z' maps isomorphically to Z' (use Divisors, Lemmas lemma strict transform (Tag 080E) and lemma blow up effective Cartier divisor (Tag 0807)). In other words, if an irreducible component Z of X_i is not contracted under the map $X_i \to X_j$ $(i > j)$ then it maps isomorphically to its image.

Now we are ready to prove the lemma. Let $E \subset Y_m$ be the exceptional curve of the first kind which is contracted by the morphism $Y_m \to Y_{m-1}$. If E is contracted by the morphism $Y_m = X_n \to X$, then there is a factorization $Y_{m-1} \to X$ (Resolution of Surfaces, Lemma lemma factor through contraction (Tag 0C5J)) and moreover $Y_{m-1} \to X$ is a sequence of blowups in closed points (Resolution of Surfaces, Lemma lemma proper birational regular surfaces (Tag 0C5R)). In this case we lower m and we win by induction. Finally, assume that E is not contracted by the morphism $Y_m \to X$. Then $E \to X_k$ is surjective as X_k is irreducible and by the above this means it is an isomorphism. Hence X_k is isomorphic to a projective line as desired. $\square$